

CS 121 Assignment 3: Milestone 3

Search Engine Report

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Overview

For M3 we ran 20 test queries to see how well our search engine was doing. About half of them gave bad results so we made some changes to fix those. We also added a few extra credit features on top of what we had for M2.

Improvements Made

Cosine Normalization

Longer pages were getting way higher scores just because they had more words. We divided each score by the document length to fix that.

URL Boosting

If a query term shows up in the URL itself, that page is probably more relevant. We give a +3.0 bonus per query term found in the URL.

URL Penalty System

Some URL patterns kept showing up with bad results like old slide pages, .txt files, and raw archive dataset folders. We added score penalties for those.

Partial Match Fallback

If AND gives zero results because a term is too rare, we fall back to the two rarest terms and return partial matches instead of nothing.

2-gram Indexing

We index pairs of consecutive words as bigrams during indexing. At search time, if the query bigram is found in a page, that page gets a 1.5x score bonus. Helps a lot with multi-word queries.

Extra Credit Features

Feature	Description
Exact + Near Duplicate Detection	MD5 for exact duplicates. SimHash with Hamming distance for near-duplicates (threshold: 4 bits). Both in indexer.py.
2-gram Indexing	Bigrams indexed during construction, used at search time with a 1.5x score bonus.
Web Interface	Flask web interface (search_web.py). Run with 'py search_web.py', open http://localhost:5000 .

Queries That Performed Well

Query	Notes	Time
cristina lopes	Profile page ranked #1, URL boost helped	2ms
ACM	Relevant pages, bibliography files removed by penalty	17ms
python programming	Python tutorial pages ranked correctly	11ms
artificial intelligence	AI research and course pages returned	9ms
graph algorithms	Eppstein graph page ranked #1	11ms
neural network	Relevant research pages returned	5ms
natural language processing	NLP course and research pages returned	17ms
undergraduate admissions	Admissions page clearly #1 after URL boost	12ms
research paper	Faculty research pages ranked well	43ms
software engineering jobs	Job search and career pages returned	22ms

Queries That Performed Poorly

These 10 queries gave bad results. We explain what was wrong and what we changed.

"machine learning"

Problem: All results were raw dataset folders from archive.ics.uci.edu, not actual ML pages.

Fix: Added a penalty for those archive URLs so they stop showing up at the top.

"data structures"

Problem: Old NASA slide pages kept ranking #1, nothing about actual CS data structures.

Fix: Added penalty for /tsld slide paths and archive pages. Cosine norm helped too.

"master of software engineering"

Problem: General software engineering pages showed up instead of the mswe.ics.uci.edu pages.

Fix: URL boost helped push pages with those terms in the URL higher.

"web crawler"

Problem: Results were mostly random ICS pages. The word 'crawler' is pretty rare in the index.

Fix: Partial match fallback kicks in and returns the best partial matches instead of nothing.

"natural language processing"

Problem: A .txt file ranked #1 which is not a useful result at all.

Fix: Added a penalty for .txt URLs to push those down.

"faculty office hours"

Problem: Just got generic grad school pages, nothing about office hours specifically.

Fix: Cosine norm helped reduce the advantage of long pages that just happen to have those words.

"ICS course schedule"

Problem: Only got pages from one professor's ics-6a class, not a general schedule.

Fix: Cosine norm reduced those pages' advantage. This one is hard since there is no single schedule page in the corpus.

"operating systems"

Problem: A random auto-generated slide page ranked #1 instead of the OS course page.

Fix: Added penalty for /sld paths. The ics143 course page now ranks higher.

"database management"

Problem: A .txt file was ranking at the top, not useful at all.

Fix: Penalty for .txt URLs moved it down. Shorter, more focused pages ranked higher after cosine norm.

"computer science UCI"

Problem: Random contact pages ranked above the actual CS department page.

Fix: URL boost pushed cs.uci.edu pages up. CS grad programs page now in top 5.